

SAN VALENTINO ALLA MUTA, Italy

WMO: 160080

Lat: 46.7619N

Lon: 10.5344E

Elev: 4793

StdP: 12.32

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.3	10.9	-3.8	5.3	14.3	-0.2	6.5	18.1	19.5	19.3	16.8	21.6	5.5	10	0.599

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.3	76.9	57.8	73.6	57.0	70.8	56.3	61.9	70.4	60.3	67.9	58.9	66.3	5.5	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.2	89.9	64.4	57.5	84.5	63.0	56.1	80.1	61.7	30.2	70.7	29.0	68.0	27.9	66.2	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.7	13.1	11.4	3.2	82.2	5.1	3.7	-0.5	84.9	-3.4	87.0	-6.3	89.1	-10.0	91.8		
(5)			2.4	65.6	5.4	3.2	-1.4	67.9	-4.5	69.7	-7.5	71.5	-11.4	73.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.5	26.5	28.4	35.6	42.0	49.5	56.6	59.6	59.0	52.6	45.7	36.2	29.2		
	DBStd	13.31	6.84	7.54	7.13	6.42	5.60	6.30	5.38	5.11	5.45	6.34	7.03	7.05		
	HDD50	3381	728	605	447	249	77	15	3	3	34	158	416	646		
	HDD65	7873	1193	1025	912	690	480	257	178	194	371	597	865	1111		
	CDD50	1011	0	0	0	9	62	214	302	283	113	26	2	0		
	CDD65	27	0	0	0	0	0	6	12	9	0	0	0	0		
	CDH74	257	0	0	0	0	1	76	97	78	5	0	0	0		
	CDH80	29	0	0	0	0	0	9	9	10	1	0	0	0		
	WSAvg	4.6	4.6	4.9	4.7	5.1	4.8	4.5	4.4	4.0	4.4	4.4	4.5	4.6		
	PrecAvg	27.6	1.4	1.3	1.5	1.5	2.4	3.3	4.1	4.1	2.6	2.3	2.0	1.5		
(15) Precipitation	PrecMax	35.3	3.3	6.3	3.7	3.1	4.8	4.9	6.5	7.4	5.5	6.1	6.2	4.2		
	PrecMin	20.0	0.3	0.3	0.3	0.3	0.7	1.5	2.8	1.7	0.9	0.0	0.2	0.2		
	PrecStd	3.9	0.8	1.2	0.8	0.6	0.9	0.9	0.9	1.4	1.2	1.3	1.3	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.8	51.5	59.4	67.2	71.8	80.4	81.0	80.8	72.2	66.4	58.2	48.5		
		MCWB	37.3	38.1	42.0	49.0	53.5	57.4	60.4	59.2	54.8	50.2	46.8	37.7		
	2%	DB	41.4	46.0	54.5	61.4	67.3	75.7	76.7	75.6	68.6	61.2	51.9	43.7		
		MCWB	34.6	36.1	40.7	45.8	51.0	56.7	58.2	57.9	54.4	49.3	43.0	35.7		
	5%	DB	38.0	41.5	50.0	57.1	63.6	72.0	73.4	71.9	65.2	57.9	47.8	40.0		
		MCWB	32.6	33.4	38.5	43.7	49.6	56.1	57.5	57.4	53.4	48.4	40.9	34.2		
	10%	DB	35.1	37.8	45.9	53.2	60.3	68.2	70.2	68.6	61.5	54.7	44.3	37.3		
		MCWB	30.8	31.7	36.9	42.0	48.5	54.8	56.9	56.6	52.2	47.8	39.8	32.9		
	0.4%	WB	39.3	39.9	44.6	50.8	57.0	62.9	65.3	63.5	59.9	54.8	48.9	40.0		
		MCDB	44.5	48.6	55.4	61.2	65.5	71.5	73.8	73.2	67.4	60.3	54.5	43.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.8	36.8	41.9	48.0	54.2	60.7	62.3	61.2	57.4	52.3	44.8	37.5		
		MCDB	39.7	44.4	51.3	57.5	61.9	69.3	70.9	68.6	64.1	57.2	49.1	41.2		
	5%	WB	33.4	34.4	39.7	45.5	52.1	58.5	60.5	59.8	55.6	50.5	42.3	35.2		
		MCDB	36.8	40.1	47.7	53.9	59.5	67.0	68.1	66.9	60.8	55.0	45.6	38.9		
	10%	WB	31.6	32.4	37.7	43.5	50.3	56.7	59.0	58.4	54.1	48.5	40.5	33.2		
		MCDB	34.6	36.7	44.3	50.6	56.9	64.7	66.3	65.4	59.2	53.2	43.5	36.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.3	15.0	16.4	16.3	15.7	17.1	17.3	15.9	15.0	13.9	11.3	11.8		
		MCDBR	16.2	20.0	22.9	23.4	22.0	22.9	22.5	21.7	20.5	19.5	17.2	15.0		
		MCWBR	10.8	12.1	11.6	10.3	9.1	8.7	8.5	8.2	8.7	9.6	9.9	9.5		
		MCDBR	14.3	18.1	20.5	20.3	17.9	19.5	18.8	17.1	16.7	15.4	13.8	13.4		
	5% WB	MCWBR	10.3	11.6	11.3	10.3	9.2	9.3	9.4	8.5	8.7	9.0	9.0	9.0		
(40) Clear-Sky Solar Irradiance	taub		0.242	0.269	0.316	0.366	0.381	0.388	0.384	0.377	0.341	0.315	0.276	0.243		
	taud		2.330	2.259	2.199	2.189	2.243	2.275	2.289	2.329	2.413	2.456	2.455	2.380		
	Ebn at Noon		276	287	287	281	280	278	277	275	277	268	260	263		
	Edh at Noon		23	30	38	43	42	41	40	37	31	25	21	19		
(44) All-Sky Solar Radiation	RadAvg		496	797	1177	1417	1549	1763	1775	1553	1194	806	487	410		
	RadStd		55	91	140	194	169	144	139	147	124	101	78	51		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	+1.15	N/A	N/A	N/A	N/A
(47)	Regional (107 neighbors)	+0.61	N/A	N/A	+1.13	+0.72	+0.61	N/A	N/A	+108	+27

Nomenclature: See separate page